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WHEN BOTS EVALUATE HUMANS:
DELEGATION TO BOTS AND THE RESHAPING OF AUTHORITY

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ABSTRACT

Information systems scholars typically frame the delegation of tasks to AI-based bots as a means of improving efficiency and supporting decision-making. Yet when evaluative tasks are delegated to supervisory bots, authority is transferred to them as well. This study examines how such authority, once conferred on bots, is contested and redistributed by the humans whose work is subject to the bot's evaluation. We investigate this process in the context of Wikipedia, an online peer-production community in which an antivandalism bot was given the task of autonomously reverting vandalized, i.e., maliciously edited, contributions. Drawing on the concept of performative authority, we traced how community members negotiated the enactment of the bot's newfound authority with the bot's developers through authority-negotiation design moves. We found that these moves were the outcome of a recursive process marked by the redistribution of authority across actors in decisions on vandalism, the institutionalization of structures that shaped how authority was influenced, and the evolving actions of the bot itself, which triggered renewed contestation over time. Within the context of a peer-production community, our study offers fresh insight into how authority delegated to bots evolves as a dynamic cocreation process after delegation and how humans reclaim authority once it has been ceded to a supervisory bot. We shed light on how humans preserve authority, professional discretion, and the meaningfulness of their work in AI-mediated settings.

Keywords: human-bot collaboration, delegation, bots, authority, grounded theory

WHEN BOTS EVALUATE HUMANS: DELEGATION TO BOTS AND THE RESHAPING OF AUTHORITY

INTRODUCTION

Intelligent technologies in the form of AI-based bots are being widely deployed in organizational settings. These bots are not merely tools; rather, they are capable of learning from data and acting autonomously, that is, without continuous human oversight (Berente et al., 2021). Such bots have been deployed and studied across contexts, from content moderation on social media (de Lima Salge et al., 2022) to physical robots in warehouse operations (You & Robert, 2024). Their ability to make decisions rapidly, enhance productivity, and operate at scale has encouraged widespread adoption. In a 2024 survey, 78% of organizations reported using AI in at least one function (McKinsey & Company, 2025).

While many deployments of AI-based bots target productivity (e.g., Anthony et al., 2023; Dennis et al., 2023; Sachin & Schechter, 2024), organizations are increasingly using a specific type of AI-based bot to govern human work (Berente et al., 2021; Safadi et al., 2024): namely, supervisory bots. These bots monitor goal progress and make corrections autonomously over time (Baird & Maruping, 2021). Their use is qualitatively different from using bots to automate tasks (Baird & Maruping, 2021) because they are given the authority to evaluate, intervene, and sometimes even overrule human work (Safadi et al., 2024). One such example is Wikipedia, where supervisory bots evaluate peer-produced articles and can even revert edits as part of the platform's content moderation process.

Evidence from research and practice suggests that humans often contest the authority vested in supervisory bots (Mindel et al., 2024) because authority is not a neutral feature of organizational life but one that shapes identity, status, and meaning for those who hold it or

are subject to it (Richardson et al., 2021; Vough, 2012). Since supervisory bots influence outcomes that are important for people's work, the authority they exercise affects how both work and evaluations are experienced and negotiated. As a result, humans may ignore, bypass, or question bot decisions that evaluate their work (Stelmaszak et al., 2024; Möhlmann et al., 2021).

Information systems (IS) literature on supervisory bots has so far largely centered either on the initial delegation decision itself or how humans adapt their work once bots evaluate it (e.g., Fügener et al., 2022; Jussupow et al., 2021). For instance, Dennis et al. (2023) examined how humans adapt their behavior when collaborating with AI-based bots who are introduced as team members and granted decision rights for work-related outcomes. This line of research treats the authority of bots as a stable feature of the work arrangement, which is granted or withheld on fixed terms at the point of delegation.

However, other research suggests that with AI-based bots, authority is empirically distinct from delegation (Demetis & Lee, 2018). Whereas delegation implies a decision to grant decision rights at a given point, by contrast, authority granted to AI-based bots continuously evolves after delegation because bots learn from data and adjust how they evaluate and act. This can have important consequences for human discretion in decision-making, potentially impacting individuals' behavioral choices, shaping decision outcomes through data-driven biases, and limiting individuals' ability to gain power and influence—effects which may compound as bot learning continues (Kane et al., 2020).

Prior work also shows that when technologies constrain humans' decision-making capabilities, humans often push back and attempt to reclaim authority (Miranda et al., 2016; Young et al., 2021). However, we currently lack adequate theory regarding how, precisely, humans without formal positions of power, those who did not get but are subject to bot authority, can reclaim authority that has been vested in bots and then evolves in practice. This

necessitates deeper theorizing to understand how humans navigate this authority and wrest it back once it has been ceded to supervisory bots, and how these postdelegation processes reshapes the bot's authority over time. This motivates our core research question: *How do individuals subject to supervisory bots contest and reclaim authority in practice, and how does this affect the evaluation of who wields authority?*

We explore this question in the context of vandalism on Wikipedia, defined as “editing a page in an intentionally disruptive or malicious manner” (“Vandalism on Wikipedia,” 2025). Wikipedia uses a supervisory bot called ClueBot NG (hereafter, ClueBot) to autonomously detect and revert edits it deems as vandalism, tasks which were previously performed by community members (CMs). However, the bot can mistakenly override legitimate contributions or fail to flag problematic ones, triggering negotiation between the developers who developed and maintain the bot and the CMs affected by its actions (Mindel et al., 2024; Nguyen et al., 2024). Because Wikipedia promotes participatory and transparent governance in peer production, these negotiations are publicly documented and reflected in subsequent code changes, providing a rare instance of traceability in how authority is reshaped over time.

We analyze these negotiations through the lens of performative authority, which views authority as being enacted in everyday practice (Bourgoin et al., 2020). Our analysis used a grounded theory approach to show that CMs engaged in a series of *authority-negotiation design moves* with developers to reshape and reclaim authority ceded to the bot. Our findings show that each move was instantiated through changes to the bot's design, for instance, modifying detection thresholds or adding a flagging mechanism for questionable bot edits. These moves changed over time, altering how authority was enacted by the bot in practice, and also created new structures to enable CMs to contest the bot in new and more sustained ways, which allowed them to gradually reclaim authority in the content moderation process.

Through analysis of this case study, our work contributes to the understanding of how authority instantiated in code and delegated to a supervisory bot continues to evolve through dynamic and recursive cocreation processes in a peer production community. We demonstrate that authority can be reshaped through authority-negotiation design moves that both redistribute influence across stakeholders, such as CMs, developers, and the bot, and reconfigure the structures and scope through which authority is exercised. In doing so, we extend the IS research on delegation by shifting attention from the delegation decisions themselves to the ongoing practices through which humans navigate and adjust the authority of supervisory bots in work arrangements. Conceptually, we show that the locus of evolving authority does not solely lie with the bot itself; rather, it is instantiated in the practices of those subject to bot authority and reflected in how they negotiate its enactment through interactions with developers and design change requests. While our analysis centers on a peer-production context where authority is co-constructed, the authority-negotiation design moves we identify are increasingly salient because similar evaluative and supervisory roles are being delegated to contemporary AI systems ranging from decision support to generative models. Practically, our findings highlight that delegating authority to bots to evaluate human work is not a one-time decision but an ongoing process that requires structures for negotiation and adjustment.

BACKGROUND

This section provides a brief overview of what is currently known about the delegation of tasks to AI-based bots and the implications for authority. We then discuss the concepts of performative authority, influence, and design moves, which provide an initial conceptual scaffolding for our analysis.

Delegation to Supervisory Bots and Authority

The IS literature defines *delegation* as “transferring rights and responsibilities for task

execution and outcomes to another” (Baird & Maruping, 2021, p. 317). However, delegation extends beyond the task allocation; rather, it involves ceding authority and granting the other party autonomy (Leana, 1987). With the emergence of bots capable of autonomous action without human oversight (Berente et al., 2021; Safadi et al., 2024), recent research has begun to focus on how delegation extends from humans to supervisory bots, which is consequential because humans give supervisory bots the authority to manage, supervise, and influence human activity (Safadi et al., 2024).

Authority is the capacity for action constitutive of agency (Haugaard, 2017) and entails the ability to decide and make choices regarding decision outcomes (Leanna, 1987). It can be understood both in the terms of its scope (what authority entails—its range of actions and decisions) and its grounds (what constitutes authority—its basis and legitimacy) (Peabody, 1962; Fleming and Spicer, 2014). Most research in IS and organizational science focuses on the latter, examining how humans evaluate a bot’s grounds for authority (e.g., its capability to solve a task) relative to their own ability to decide whether to delegate authority and how to incorporate bot outputs into their work (Jussupow et al., 2021; Lebovitz et al., 2022). Most prior studies treat delegation as the central concept, with authority relegated to an allocation of decision rights that are granted to or withheld from a bot at the point of delegation and altered only when it is explicitly revoked (Baird & Maruping, 2021; Liu et al, 2025).

Some work has examined contexts where tasks are transferred to a bot, and those subject to the bot’s authority are no longer responsible for task delegation. In such cases, autonomous bot authority, which may evolve differently from how it was originally granted, becomes the central concept instead of delegation. Bots that act autonomously learn from behavior in their environment, which can shift the scope of their authority over time and may lead to unintended consequences that would have likely been prevented by human discretion (Anthony et al., 2023; von Krogh, 2018)—for instance, the development of biases against

certain subgroups (Young et al., 2021). Indeed, scholars have warned of potential negative effects on the workforce, labor, and consumers, in terms of privacy, fairness, and discrimination if bots are allowed to act without human oversight (Berente et al., 2021; Schuetz & Venkatesh, 2020).

IS research has examined how to prevent such negative consequences. For example, Teodorescu et al. (2021) proposed mutual augmentation as a way for humans and AI to work together to avoid biases in decision-making. Also, Kane et al. (2020) outlined design principles that embed rationality, inclusion, and human agency into AI systems to guard against potential discrimination. Collectively, this work focuses on understanding and shaping the scope of bot authority at the time of development, to prevent it from evolving in unintended directions after delegation.

However, we still know relatively little about how autonomous bot authority is shaped in practice by those subject to it after delegation. Research shows that humans may collectively challenge the authority vested in technologies they perceive as limiting their discretion, even if they don't hold formal positions of power (Young, 2017; Mindel et al., 2024). Related work has highlighted the dual role of information systems—as instruments that constrain human agency but also as resources that individuals can use to counter such constraints (Miranda et al., 2016). However, this work primarily concerns technologies whose effects are set at the time of design and use, rather than systems that act autonomously and continuously learn and evolve. What remains underexplored is how autonomous bot authority evolves after it is delegated and how it is shaped in practice by those subject to it.

Conceptual Development

To fill this research gap, we bring together four concepts: *authority*, *influence*, *structures*, and *design moves*. We approach these through the lens of performative authority, which highlights how authority is continually reshaped through everyday practices (Bourgoin

et al., 2020). Together, these concepts provide a scaffolding for tracing the dynamic process through which authority delegated to bots is enacted in practice, how influence shifts across actors, and how structures emerge that sustain or constrain authority over time.

Research in organizational science shows that *authority*, i.e., the ability to make decisions that affect others (Leana, 1987), takes shape through judgments on the grounds of authority and responses to it by both delegators and delegees (Klein et al., 2006). Authority is therefore provisional rather than fixed and is continually open to reconfiguration in practice (Hamman & Martinez-Carrasco, 2023; Dosi et al., 2021; Hanisch et al., 2024). This may occur explicitly—for example, when authority is withdrawn by delegators (Hanisch et al., 2024), but more often it is done tacitly through everyday practices that change how authority is enacted by those subject to it (Fehrler & Janas, 2021; Huising, 2015). For example, Vásquez et al. (2018) showed how organizational members who voice and collectively negotiate concerns about authority can form alliances that reshape authority and grant standing to subordinates who would otherwise lack formal power. Similarly, Wellman (2017) emphasized that such shifts in authority hinge on the mobilization of *influence*, as authority is reshaped through interactions that enable those without formal authority to gain influence over how others make decisions.

Research shows how the tacit shaping of authority builds on the concept of performativity, highlighting how practices constitute new realities (Garud et al., 2018). These practices, in turn, unfold within organizational *structures* in the form of rules and technological architectures that define both how authority can be exercised and how those subject to it can gain influence over outcomes (Feldman & Pentland, 2003; Giddens, 1984). As Lee (2024) explain, authority emerges from the interplay between formal structures, such as governance, that sustain and constrain authority and the practices that challenge it. However, structures are not static constraints but are reshaped as authority itself is reconfigured, giving rise to

new structures that enable influence. Within these bounds, those subject to authority can resist and reshape it in organizationally permissible ways (Fleming & Spicer, 2014). Contesting authority therefore rarely manifests as a wholesale revolt against authority. Instead, it is shaped through incremental acts of defiance, which accumulate over time to reconfigure authority and create structures that further influence how authority evolves (Ewick & Silbey, 2003).

While there is ample research on how performative authority is enacted and reshaped in human contexts, bot authority differs in two important ways. First, the actor performing authority is non-human, and the scope and grounds for its authority are initially defined by human developers who delegate tasks to the bot. As a result, the responses of those subject to the bot's authority are directed not at the bot but at the developers vesting the bot with authority in the first place. Second, bots cannot flexibly adjust how they exercise authority in situ in response to contestation of those subject to it. Their authority is performed in accordance with the design specified by the developers, making it immune to social processes of adaptation. Hence, any changes must materialize in bot design. We therefore use the concept of *design moves*—defined as discrete, strategic changes to artifacts that, over time, accumulate into structures that shape future action (Woodard et al., 2013)—to capture how the responses of those subject to the bot's authority materialize in discrete changes to a bot's design that then shape its authority. Table 1 summarizes the four core concepts.

Table 1. *Sensitizing concepts and their description*

Sensitizing concepts	Description	Informing sources
Authority	The ability to make decisions that affect others, including determining outcomes on their behalf.	Leana (1987)
Influence	The capacity to shape outcomes through interactions that affect how others make decisions.	Wellman (2017)
Structures	Rules and resources that both enable and constrain how authority is enacted.	Giddens (1984)
Design moves	Discrete, strategic changes to artifacts that, over time, accumulate into structures that shape future action.	Woodard et al., (2013)

In our study, we use *authority* to examine how autonomous bot authority is shaped by those subject to it, *influence* to trace how actors gain standing in shaping authority, *structures* to analyze how authority is sustained or constrained through organizational and technical arrangements, and *design moves* to capture how the responses of those subject to bots' authority materialize in changes to bot design. These concepts provide the analytical framework that guides our research design.

RESEARCH DESIGN

To situate our study, we provide an overview of the context examined by our study. We first outline the context of peer production and the role of bots within it. Then, we illustrate Wikipedia's governance structures that shape how bots are developed and maintained. Finally, we describe the formal approval process that regulates how bots become active within the community. We then provide an overview of the data and explain our three-stage approach based on grounded theory that we used to draw insights from the interactions among CMs, bot developers, and ClueBot on Wikipedia.

Research Context

Peer Production Communities and the Role of Bots in Quality Control

Open peer production communities, such as Wikipedia, consist of loosely connected CMs who make contributions through an internet-based platform (Kane et al., 2014). These communities are open to the public and allow CMs to pursue their own objectives and contribute voluntarily within the boundaries of the platform (O'Mahony & Ferraro, 2007). For instance, on Wikipedia, CMs can engage in activities such as writing and editing encyclopedia-like articles, performing administrative roles, and ensuring quality control (Arazy et al., 2015).

On Wikipedia, voluntary contributions result in more than two article edits per second

(“Wikipedia:Statistics,” 2024). However, 7% of all edits to Wikipedia constitute *vandalism*, that is, deliberate attempts to compromise the integrity of articles (Geiger & Halfaker, 2013) by inserting spam, false information, or profanity (Tramullas et al., 2016). To combat vandalism, Wikipedia employs bots such as ClueBot to monitor articles and revert inappropriate edits (“Wikipedia:Bots,” 2024; Zheng et al., 2019)

ClueBot is Wikipedia’s most prolific antivandalism bot, capturing around 70% of the platform’s instances of vandalism (Breneman, 2010). ClueBot combines two machine-learning algorithms—a naïve Bayes classifier and an artificial neural network (ANN)—to detect and revert vandalism. The classifier estimates the probability of vandalism based on word patterns in both vandalistic and non-vandalistic edits, whereas the ANN improves detection accuracy by capturing more complex patterns and relationships (Breneman, 2010). These algorithms enable ClueBot to operate in real time, scanning and reverting vandalism without human oversight (Geiger & Halfaker, 2013).

The evaluative task of detecting and reverting vandalism was delegated to bots such as ClueBot to complement rather than replace the voluntary efforts of CMs who also combat vandalism. As a result, both human editors and bots participate in maintaining Wikipedia’s quality standards.

Governance Structures on Wikipedia

As both humans and bots are responsible for detecting and reverting instances of vandalism, tensions can arise when the bot’s reversion contradicts human edits, raising questions about editorial discretion and oversight. Figure 1 illustrates such a case: ClueBot (shown on the right) reinserted vandalism into an article after user 141.219.227.209 had correctly removed it from an earlier edit by user NRTA. When the human editor attempted to override the bot, ClueBot reinstated the vandalism, leaving the editor unable to remove it. In principle, human editors may revert the bot’s edits just as they would the edits of another

user. In practice, however, this proves ineffective for the human editors, as the bot immediately reinstates what it has classified as vandalism, even when incorrect, thereby restricting the discretion of human editors.

Incidents where human editors' discretion conflicts with bot actions became visible during the formal approval process for ClueBot. Before it was deployed, ClueBot required approval through Wikipedia's bot-approval process, which included test phases. This process began with a public request posted by the bot's developers describing the bot's purpose and functionality. Other members then commented on the potential risks, unintended consequences, and the bot's alignment with editorial norms on a dedicated discussion page. For example, during ClueBot's approval, CMs debated whether the bot's detection of vandalism might interfere with legitimate edits. These exchanges informed later decisions, such as refining test parameters and excluding certain user groups from automated reversion, such as very experienced or new editors. Participation in this process was voluntary and drew from CMs active in governance spaces such as WikiProject Bots and WikiProject Counter-Vandalism.

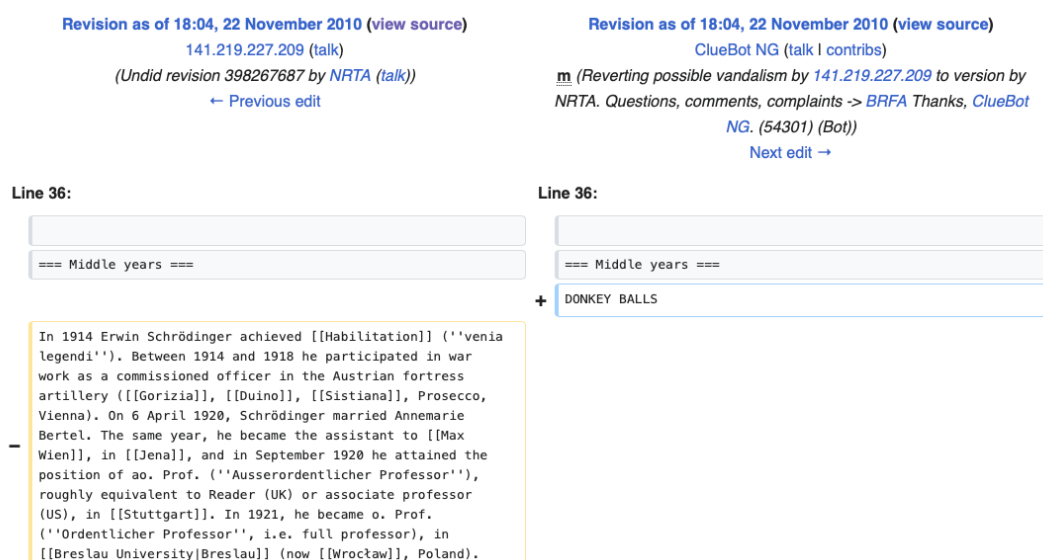


Figure 1. Example of vandalism edits by a human editor and ClueBot

Bot-Approval Process on Wikipedia

Wikipedia’s bot-approval process governs how bots such as ClueBot are proposed, tested, and approved. The bot-approval process is overseen by the Bot Approvals Group (BAG), a governing body composed of experienced CMs with technical expertise in bot development (“Wikipedia:Bots Approvals Group,” 2024). The BAG provides a formal initial review of the bot’s safety and utility before coordinating community discourse through the bot talk page and initiating pretest and test phases. These phases may span several weeks or months and are based on predefined time frames (e.g., 7- or 14-day test phases). In the case of ClueBot, they consisted of a 7-day pretrial and two 14-day trial periods before final approval. During these phases, developers are expected to address community feedback, which carries weight because developer responses are overseen by the BAG. The BAG may also suspend or extend testing based on ongoing feedback from the community, and it ultimately determines whether a bot is approved (“Wikipedia:Bots Approvals Group,” 2024). Once a bot is approved, bots remain subject to community discussion on their talk pages, although the BAG no longer monitors this feedback. In this way, the BAG provides a structure that governs how bots are introduced, tested, and maintained, and empowers the feedback of CMs by requiring developers to address it.

Table 2 provides an overview of the changes (C-n) made by developers (D_1 and D_2) during the bot-approval process for ClueBot. In total, discussions on the bot’s talk page led to 16 changes to ClueBot across the pretrial and trial phases. With one exception (C3, introduced in the pretrial phase and later replaced by C16 in the second trial phase), all changes remained in place from their introduction to the bot’s final approval.

Table 2: *Timeline and changes made to the bot during the approval process.*

Timeline of bot approval	Timestamp (UTC)	Change no.	Key actor	Adaptation to bot
Request for bot approval	25.10.2010; 00:35		D_1, D_2	
Pretrial	25.10.2010; 06:55	C-1	D_1	Discarding data on previous contribution history to avoid labeling new editors as vandals for minor early editing mistakes
	30.10.2010; 12:04	C-2	D_1	Adapting parameters to improve the ANN's false-positive scores
	30.10.2010; 21:53	C-3	D_1	Lowering the false-positive rate from 0.5% to 0.25%
Approved for trial (14 days)	02.11.2010; 04:37		BAG	
Trial 1	02.11.2010; 19:30	C-4	D_1	Excluding redirects in the wiki markup from automatic bot reversions
	02.11.2010; 20:38	C-5	D_1	Whitelisting of experienced editors with 50+ edits to avoid resentment
	03.11.2010; 00:03	C-6	D_1	Adding ambiguity tags to enable human checkups in cases of borderline vandalism
	03.11.2010; 16:31	C-7	D_1	Adding exceptions for novice editor whose content was inexplicably reverted by the bot
	03.11.2010; 16:33	C-8	D_2	Banning bot reversions of any edit in the main namespace
	03.11.2010; 17:47	C-9	D_2	Fixing the exclusion compliance for the bot
	05.11.2010; 19:40	C-10	D_2	Revising the registration message to make it more intuitive for new editors
	06.11.2010; 07:07	C-11	D_1	Adjust statistical word criteria
	08.11.2010; 12:25	C-12	D_1	Fix interface bug that mistakenly asks CMs to classify their own vandalism edits
	13.11.2010; 16:21	C-13	D_2	Including editor votes for disagreements ("vandalism," "constructive," "skip")
	16.11.2010; 16:21	C-14	D_2	Making other editors' previous votes visible and adding space for comments to allow consensus on decisions of borderline edits to emerge
Approved for extended trial (14 days)	18.11.2010; 04:40		BAG	
Trial 2	22.11.2010; 19:08	C-15	D_2	Fixing multiple headers in the interface
	29.11.2010; 03:34	C-16	D_1	Lowering the false-positive rate from 0.25% to 0.1%
Approved to operate at operator's discretion	03.12.2010; 02:24		BAG	

Taken together, the bot-approval process of ClueBot provides a unique and transparent context for studying how bots are integrated into the work of a peer production community and how their scope and responsibilities are adapted by those subject to them. ClueBot's

approval process and the open discussions on talk pages offer a clear window into how bots operate in practice and how stakeholders can advocate for changes to their operation.

Data Collection

Our data-collection approach began with the ClueBot’s project page (“Wikipedia: Bots/Requests for approval/ClueBot,” 2024), which contains the developers’ formal submission for bot approval to the BAG, including details such as source code availability, programming language, and functionality. It also documents the BAG’s decisions during the approval process, which included two trial periods and the final bot approval. The project page further archives associated talk page discussions, where CMs provided comments or raised concerns and developers were required to respond, either by explaining the bot’s behavior or by implementing changes to the bot’s design, task execution, or scope. Together, these materials allowed us to reconstruct the timeline for ClueBot’s approval process and trace changes implemented by developers, from the original bot approval request on October 25, 2010, to the bot’s final approval on December 3, 2010, as shown in Table 2.

We then traced additional discussions referenced on the talk page, such as links to related issues raised elsewhere on Wikipedia by CMs and brought to the developers’ attention on the bot’s talk page. This led us to collect material from two further sources: a dedicated page for reporting bot false positives and archived discussions from the ClueBot Commons page (“User talk: ClueBot Commons/Userpage,” 2024), which hosted a combined discussion section for ClueBot III (ClueBot’s predecessor) and ClueBot. From these pages, we extracted all discussions between CMs and developers that concerned ClueBot posted between October and December 2010; we excluded discussion threads unrelated to ClueBot.

This yielded a dataset containing CM and developer discourse covering the entire formal bot approval process, from the initial request to the BAG’s final authorization of the bot. We excluded postapproval feedback to maintain a clear analytical boundary around the formal

approval process, where CMs advocated for changes made by developers under the BAG's oversight. This bracketing follows Langley (1999), who recommended structuring process analyses around meaningful temporal phases, and Eisenhardt (1989), who emphasized that imposing boundaries around cases preserves the internal coherence of an analysis.

Figure 2 provides an overview of these data sources, including descriptions of their key characteristics and explanations of how our study used them. The dataset descriptions include the number of discussion threads we collected as well as the CMs involved in the discussions. We distinguished between the CMs who originally developed and delegated tasks to ClueBot (i.e., developers) and those who commented on the talk pages (i.e., CMs). Thus, although developers were technically also CMs, we treated them as a distinct actor group because they held decision rights over the bot's initial design and implementation, whereas other CMs engaged mainly through feedback and change requests and did not have formal decision rights. Our dataset comprises 104 discussion threads (i.e., sequences of messages between CMs and developers on an original topic) and 26 additional threads based on false-positive reports, yielding 130 threads overall. We are confident that these threads provide a thorough representation and a complete set of the discussions between CMs and developers during the bot approval process because Wikipedia's editorial norms require dispersed conversations to be redirected to the central project or talk pages, which consolidates all relevant discussions in a single location. Altogether, the dataset amounted to 43,351 words.

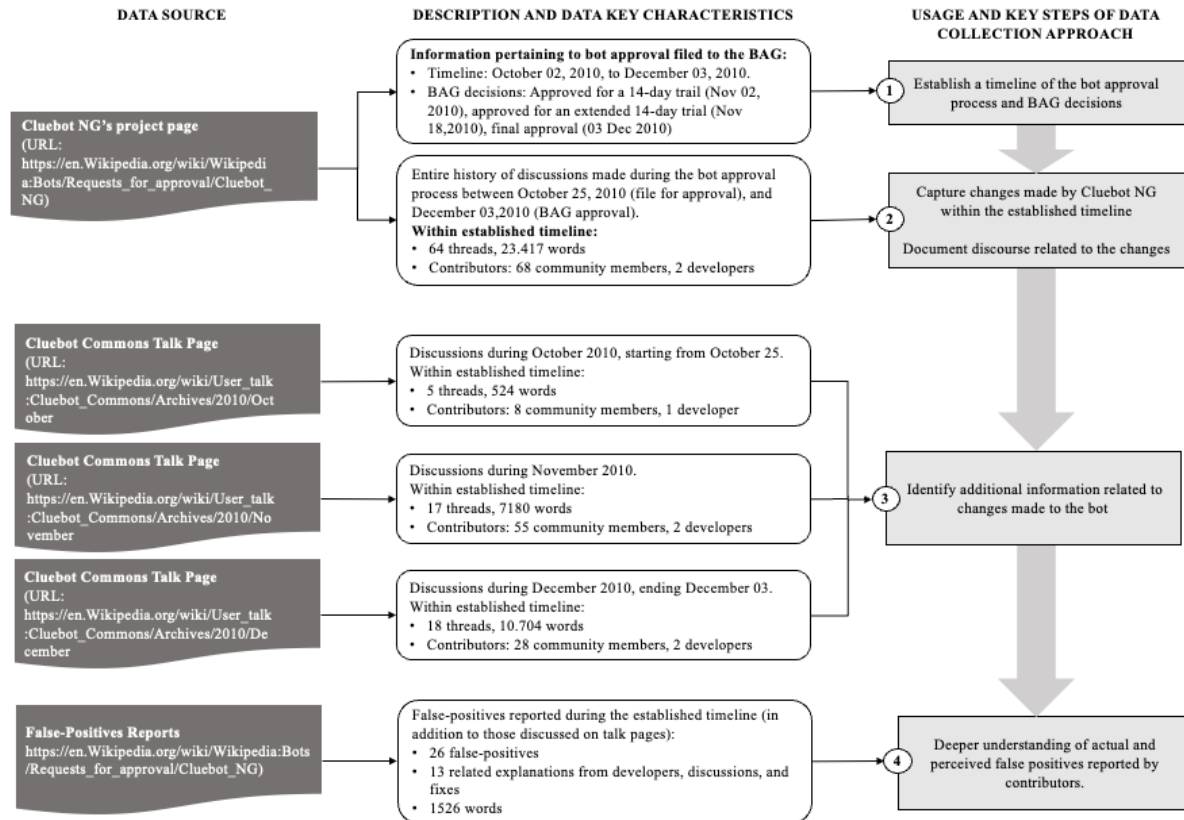


Figure 2: Descriptions of data sources and their use in the study.

Data Analysis

We adopted grounded theory as our overall logic of inquiry (Glaser & Strauss, 1967) to guide all the steps of our analysis. Following Langley's (1999) recommendation, we treated grounded theory not as a method for a discrete phase of the study but as an interpretative strategy to make sense of our rich, multilayered process-related dataset. Coding unfolded in three iterative steps, each comprising multiple stages of analysis. First, we identified and coded design-related adaptations as authority-negotiation design moves. Second, we examined how these moves produced different structures of authority. Third, we traced how these structures evolved over time to reveal changing patterns of influence among actors.

The *first step* of our analysis involved three stages of coding: open, axial, and selective. In this step, we used the concept of design moves as a sensitizing concept to code discussions between CMs and developers. Specifically, we used in vivo open coding to examine the

reasons that motivated CMs to advocate for changes in order to reveal the intent behind them and the ways they were expected to shape the bot's future evaluative work. We coded text passages if they contained (1) suggestions for adaptations (e.g., suggestions for lowering the false-positive rate from 0.25% to 0.1%); (2) arguments, problematizations, or explanations that provided a justification for adaptations (e.g., concerns that the 0.25% false-positive rate might discourage new editors, concerns about the need to lower it, or developers' rationale that the rate would improve over time); or (3) discussions of the consequences of adaptations (e.g., debates about ClueBot's vandalism false-positive rate dropping below 0.1%). This coding produced 120 open codes capturing motives and expected effects of bot adaptations.

Next, we conducted axial coding to systematically summarize and categorize these open codes. For example, multiple codes captured CMs' efforts to limit the bot's reach by exempting certain contributor groups, while others reflected attempts to create mechanisms for signaling ambiguous or disputed bot reversions to developers beyond the approval process. Axial codes represent recurring logics in how CMs sought to shape the bot's evaluative authority.

Finally, through selective coding, we consolidated the axial categories into higher-order concepts that represented distinct ways of negotiating the bot's authority. We term these *authority-negotiation design moves* because they describe patterned ways in which CMs influenced how authority was enacted by the bot. We identified five such moves: scoping, adapting, flagging, escalating, and requesting. For instance, *scoping* involved limiting the bot's authority based on CM type or contribution history (e.g., exempting newcomers and trusted, experienced editors from bot oversight). We summarize the changes, the authority-negotiation design moves, and the role of actors (CMs, developers, and BAG) in Figure 2.

After a first pass at coding the data, in the *second step*, we examined how authority-negotiation design moves created structures, that is, how CMs, developers, or ClueBot gained

or lost standing in decision-making on vandalism reversion. To this end, we used open, axial, and selective coding to recode the text passages from the previously identified authority-negotiation design moves, this time focusing not on the adaptations they produced but on how the moves altered whose authority was amplified or restrained in a sustained manner, such as through implemented rules, established procedures, or changes to the bot. From this analysis, we identified three forms of authority: codified, mediated, and direct authority.

For example, axial codes related to codified authority included *the bot's authority scaled back through rule codification* and *the bot's authority adapted through parameter codification*. These codes reflected cases where CMs negotiated bot actions they perceived to be misaligned with community values or demotivating to newcomers, leading developers to recode the bot's parameters or rules for determining vandalism. In this way, the bot's authority was restricted through developers' implementation of authority-negotiation design moves, which granted CMs greater discretion in their edits and safeguarded contributions from certain groups. In contrast, axial codes related to mediated authority, such as *rebalancing authority through community-based evaluations* or *requesting changes from developers*, showed that CMs could gain authority beyond negotiations. Importantly, this did not alter the bot's overall functioning but was exercised on a case-by-case basis. CMs could flag or vote on questionable reversions, which prompted developer intervention. In such cases, the bot's authority was scaled back, developer authority was temporarily activated, and CMs were able to shape individual outcomes without altering the bot's general functioning.

Eventually, in the *third step*, we examined how codified, mediated, and direct authority changed the influence of actors over one another and evaluated the persistence of that influence. For instance, with direct authority, CMs implemented a structure that allowed them to exert influence over the bot's immediate actions by collectively voting to overwrite its decisions when deemed necessary. We refer to this as *continuous influence*, since CMs

gained an ongoing opportunity to intervene on a case-by-case basis whenever they saw fit, as long as they were able to reach consensus. In contrast, with mediated authority, CMs could trigger the activation of developer authority, but the final decision rested with developers rather than CMs. Thus, mediated authority gave CMs influence over developers and, indirectly, influence over the bot, while direct authority enabled them to override bot decisions themselves.

Bringing the three steps of coding together, we structured these insights using temporal bracketing (Langley, 1999), segmenting the development into analytically distinct phases, each marked by who exerted influence, how it was exercised, how persistent it was, and through which authority-negotiation design moves it materialized. This approach allowed us to trace the evolution of community influence from case-by-case changes negotiated on talk pages to the emergence of more formalized ways for CMs to shape ClueBot's authority. We consolidated these insights into three phases of authority configuration. First, *malleable adaptation* describes a phase in which developers had authority to determine whether and how to implement changes to the bot's functioning based on feedback from CMs. Second, *on-request intermediation* captures a phase in which CMs selectively endorsed developer oversight through flagging, thereby amplifying their own authority over what the bot was permitted to do. Third, *continuous influence* refers to a phase in which community voting enabled CMs to exercise direct authority over the bot's decisions, allowing them to collectively override or affirm its actions. Table 3 illustrates the coding logic and resulting codes across all three stages of analysis, showing how successive rounds of coding progressively contributed to the reshaping of authority. We elaborate these three phases in the following, where we show how each phase materialized through specific authority-negotiation design moves.

Table 3: Coding approach.

Representative quotations (parts related to open codes are underlined)	Representative open codes	Axial code	Selective code
“I <u>am not sure what the magic number [false-positive rate] is going to be</u>, but my suggestion would be that as a whole the community seems to <u>prefer dealing with vandalism manually if it avoids labeling new users as vandals.</u>” (slakr, 25 Oct.) “Also, what is the <u>number of edits</u> above which a registered <u>user will be ignored.</u>” (SoleSoul, 02 Nov.) “In response to your <u>concerns about new users being labeled as vandals</u>, I’m re-running the trial dataset right now, <u>discarding any data about previous user contributions.</u>” (Crispy1989, 25 Oct.)	Step 1: Identifying authority-negotiation design moves - CMs question edit thresholds for bot’s action. - CMs object to the bot’s reversions of newcomers’ contributions. - CMs suggest preference for human handling.	Limiting the bot’s reach based on CM type and contribution	Scoping
	Step 2: Analyzing changes to authority structures - CMs contest the bot’s authority over newcomers and experienced CMs. - Developers encode exemptions.	The bot’s authority scaled back through rule codification	Codified authority
	Step 3: Understanding shifts in influence - CMs invoke overreach risks. - CMs request to define and limit bot action domain and reach.	Negotiation-based shaping of bot influence	Malleable adaptation
“If users do indeed <u>contribute less time to vandalism reversion</u>, the more likely outcome will be a <u>deduction in their total contribution</u>, because most users with a desire to write content are already doing so.” (Peter Karlsen, 28 Nov.) “I would suggest that setting the <u>acceptable false positive rate as low as it possibly can be</u> while still reverting a reasonable portion of vandalism <u>increases the probability that the bot will be approved</u> after its trial period.” (Peter Karlsen, 26 Oct.) “We are using our best judgement to <u>adapt these parameters [Bayesian score] to the optimal values.</u>” (Crispy 1989, 30 Oct.)	Step 1: Identifying authority-negotiation design moves - CMs share concern that automation displaces valued community efforts. - CMs request that developers lower the false-positive rate to minimize demotivation of CMs as a result of automation.	Adjusting the bot’s internal parameters (thresholds, classification logic)	Adaptation
	Step 2: Analyzing changes to authority structures - CMs contest bot authority, as it conflicts with community values. - Developers adapt the bot’s parameters.	The bot’s authority adapted through parameter codification	Codified authority
	Step 3: Understanding shifts in influence - CMs invoke risk to the community. - Developer requests the authority to steer the bot’s actions.	Negotiation-based steering of the bot’s parameters	Malleable adaptation
“The robot is automatically reverting my edition on the page Ideologia but ‘Ideologia’ could refer either to the <u>album Ideologia</u> and to the <u>album Ideologia (Cazuza’s album)</u>. The robot <u>needs some adjustment.</u>” (187.68.100.92, 02 Nov.) “The problem is not that it thought redirects were vandalism, but that there was <u>no special handling for ‘redirect’</u>. This has been fixed (<u>special handling added</u>)	Step 1: Identifying authority-negotiation design moves - CMs highlight questionable or incorrect bot reversions. - CMs request options for human review of specific bot decisions.	Adding a means of signaling ambiguity or non-acceptance of the bot’s decisions	Flagging
	Step 2: Analyzing changes to authority structures - CMs seek developers’ judgment regarding the overriding or revision of the bot’s behavior. - CMs expand their own discretion (indirectly, through developers).	Developer authority activated through CMs	Mediated authority

... I also added a special handling for <u>disambiguation tags</u> .” (Crispy 1989, 03 Nov.)	Step 3: Understanding shifts in influence - CMs argue for need to signal questionable bot edits. - Developers implement mechanism to flag ambiguous bot reversions on a case-by-case basis. - CMs endorse the authority of developers.	Targeted intervention in ambiguous or disputed bot decisions	On-request intermediation
“I saw somewhere that if two reviewers disagree then the <u>edit will be dropped from the dataset</u>—surely that is a recipe for blunting the sensitivity of the bot? If that is the case then I wonder <u>what is happening to all those comments I’ve placed on difficult edits</u>. What needs to happen is for those edits to be reviewed even more carefully, and perhaps even put up for community discussions.” (Philip Trueman, 12 Nov.) “Can we establish <u>how many reviews are needed to reach production-level accuracy</u>, set a target for the review phase?” (Rjwilmsi, 13 Nov.) “For discarding edits ... we’ve decided to change this to a scheme where every edit is always <u>classified as something (Vandalism, Constructive, or Skip)</u>, and that the classification that is used must have <u>at least 3x the votes</u> as any other classification.” (Crispy1989, 13 Nov.)	Step 1: Identifying authority-negotiation design moves - CMs question dismissal of ambiguous edits. - Developers revise classification rule to require community consensus. - CMs request clarity regarding the threshold for production-level reliability.	Introducing mechanisms to collectively judge the bot’s decisions	Escalating
	Step 2: Analyzing changes to authority structures - CMs reject default bot decisions and push for community override of bot decisions. - CMs reassert their own authority. - Developers embed community-voting threshold.	Rebalancing of authority by CMs through community-based evaluations	Direct authority
	Step 3: Understanding shifts in influence - CMs propose formal procedures for review and escalation of bot decisions. - CMs shape criteria by which decisions are validated through direct voting.	Institutionalizing of CMs influence via formal escalation process for contesting bot decisions	Continuous influence
	Step 1: Identifying authority-negotiation design moves - CMs request functionalities to increase classification transparency. - CMs propose change to review interface for transparency of other CMs’ classification of edits. - Developers respond by planning to implement a feedback tool.	Requesting changes with developers	Requesting
“If there is any disagreement with another reviewer about any of my classifications then <u>I’d appreciate learning about it, if only to improve my own performance</u>.” (Philip Trueman, 06 Nov.) “For getting <u>feedback on difficult edits</u>, we’ve discussed ways to do this. ... The idea is to allow users to <u>view a list of all edits they’ve classified, that others have classified differently, and allow them to view and add more comments, and change their existing vote</u>.” (Crispy1989, 13 Nov.)	Step 2: Analyzing changes to authority structures - CMs become active in specifying collaboration, thereby increasing their own scope of influence. - Developers’ authority is limited by the co-configuration of design.	Asking for changes directly	Mediated authority
	Step 3: Understanding shifts in influence - CMs continuously propose refinements to developers. - CMs shape their interactions with the bot and with developers through interface, participation mechanisms, and system affordances.	Institutionalized metalevel or interface refinement	Continuous influence

FINDINGS

Contestation of Bot Authority and Resulting Authority-Negotiation Design Moves

Figure 3 illustrates authority-negotiation design moves in the form of scoping, adapting, flagging, escalating, and requesting, which we refer to using the synonym SAFER. We describe the design moves next and provide examples of how the negotiations unfolded as CMs raised concerns about observed or anticipated bot behaviors, thereby contesting its authority with developers by arguing for specific changes. Figure 3 also provides an overview of the key actors involved in the process and visualizes where, why, and through whose action changes occurred. We elaborate on this visualization in the following descriptions of the authority-negotiation design moves.

Scoping: The bot adaptations associated with the authority-negotiation design move of *scoping* refined the boundaries of the bot's actions. These adaptations allowed the community to redetermine which aspects of the knowledge-cocreation process the bot could engage in, including in which areas of Wikipedia it was permitted to detect and revert vandalism. CMs specified how the bot should handle edits made by experienced CMs versus those made by new CMs. We characterized six adjustments to ClueBot as scoping (C1, C4, C5, C7, C8, C9), which occurred primarily during the pretrial phase and the early part of the first trial. The purpose of these adjustments was primarily to avoid discouraging new editors as well as highly experienced editors from contributing due to frequent bot reversions of their edits, thereby mitigating disruptions to the collaborative editing process.

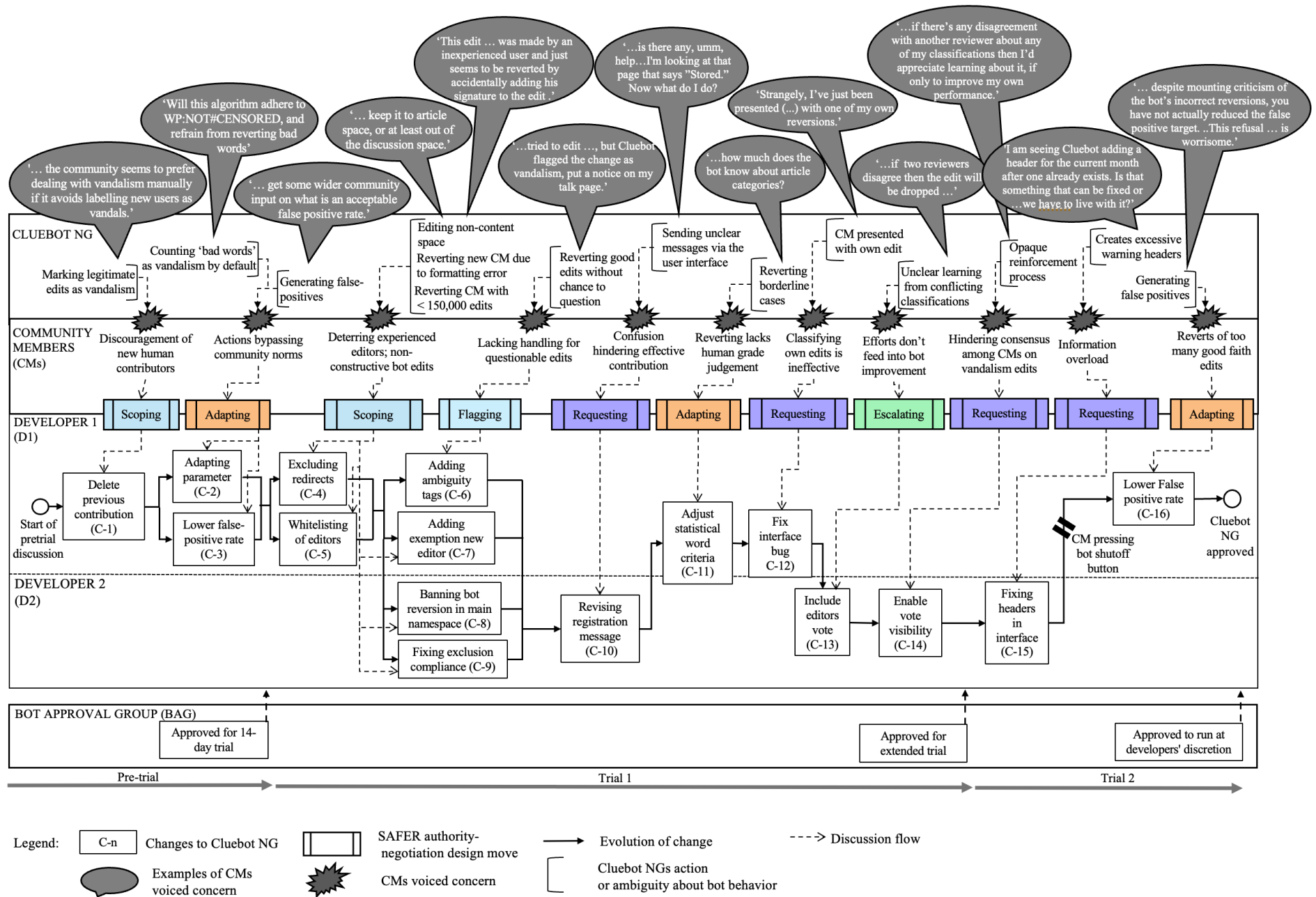


Figure 3: Visual map of authority-negotiation design moves, bot changes, and CMs contestation of bot authority.

Scoping was introduced during the pretrial phase in response to CMs' concerns that ClueBot might revert edits made by inexperienced editors more frequently than those made by seasoned editors, which could discourage new CMs and thereby undermine the community's ability to attract fresh participation. One CM stated that "the community seems to prefer dealing with vandalism manually if it avoids labeling new users as vandals" (slakr, 25 Oct.). As another member pointed out, "The community is constantly in need of new blood. ... If an experienced editor gets wrongly reverted, she presumably knows enough to take it with a grain of salt. But a good-faith user whose first or fifth contribution is reverted three seconds later by a bot is unlikely to return" (Stepheng3, 28 Nov.). In response, developers adjusted the bot's training dataset by removing information about editors; prior contribution history (Crispy1989, 25 Oct.). This change reduced the risk that editors with limited experience would be labeled as vandals because their earlier contributions contained minor mistakes.

CMs also raised concerns about how the bot's actions impacted experienced and high-volume editors. Historically, experienced editors had been granted a degree of trust; that is, their edits were typically exempt from reversions made by antivandalism bots previously used by Wikipedia. However, an incident revealed that edits made by high-volume CMs were not in fact being excluded from monitoring by ClueBot: "Since User: Tide rolls is an administrator with nearly 150,000 edits, the identification of this edit for reversion suggests that a whitelisting mechanism is not (currently) implemented" (Peter Karlsen, 26 Oct.). This led to questions such as "What is the number of edits above which a registered user will be ignored?" (Sole Soul, 2 Nov.). Developers subsequently exempted CMs who had made more than 50 edits from the bot's actions to encourage them to continue to patrol articles (Crispy1989, 2 Nov.).

Additionally, scoping limited the bot's actions to specific areas of Wikipedia. A CM noted that ClueBot was making edits beyond its designated area: "No one requested that ClueBot

reverts pages outside of the article namespace!” (Peter Karlsen, 3 Nov.). Developers responded by investigating the issue and restricting the bot’s scope to the main namespaces, thereby preventing it from making changes in areas reserved for CMs’ article drafts and internal Wikipedia processes, such as policy discussions.

Adapting allowed CMs to modify the bot’s parameters and thereby align its behavior with the community’s values and goals. In total, four adaptations (C2, C3, C11, C16) to ClueBot were associated with this authority-negotiation move, which aimed to prevent or minimize misalignment between the bot’s behavior and community norms and policies. These adjustments also aimed to bring the bot’s decision-making process into closer alignment with the way human editors were assessing vandalism on Wikipedia.

Adapting occurred in all the phases, that is, in both the pretrial phase and the two trial phases. It began when CMs expressed the concern that the parameters used to train the bot should be better aligned with Wikipedia’s policies and norms, particularly regarding content edits. For example, one CM raised the issue of censorship: “Will the algorithm of this new bot adhere to WP#CENSORED, and refrain from reverting edits solely because they contain ‘bad’ words?” (Peter Karlsen, 26 Oct.). This highlighted the concern that reverting content based on inappropriate language without considering context contradicted Wikipedia’s policy of not censoring content. In response, the developers adapted the bot’s decision-making parameters to include additional context. As one developer noted, this would allow the neural network to “learn under what circumstances certain inputs contribute to an edit being good or bad” (Crispy1989, 26 Oct.). Because the bot persisted in making inappropriate reversion, adapting led to a modification of the statistical word criteria during the first trial. Specifically, developers recognized the need to “figure out a way to determine statistical word relationship—not as in a Markov chain, or a Bayesian classifier, but in a broader sense to sort-

of automatically categorize an article” (Crispy 1989, 13 Nov.).

Adapting was also used to adjust the bot’s false-positive rate to a level deemed acceptable by the community. CMs expressed the concern that the bot might cause more harm than good if its reversions of legitimate edits outnumbered its vandalism catches: “Are you saying that if it processes 100,000 edits, 100 of which are vandalism, it will revert 70 of the vandalism edits and 500 legitimate edits? If so, that’s far worse than I thought” (Gigs, 26 Oct.). Other CMs echoed this concern, emphasizing that the acceptable false-positive rate should be no greater than that of a skilled human editor: “I would expect a percentage of false positives no greater than what an experienced, careful human editor would produce. A 0.1% false positive rate is probably at least as accurate as most humans” (Peter Karlsen, 26 Oct.).

Although adaptation efforts began during the pretrial phase and the first trial, discussions about the bot’s false-positive rate became particularly heated during the second trial. Skepticism arose regarding the developers’ calculation of false positives: “Once the false positive rate is provided in a comprehensible format, I believe the ugly truth will become apparent: one would be hard-pressed to find nearly as many false positives in 250 edits reverted by an experienced, skilled human user as there would be in the same number of edits reverted by this bot” (Peter Karlsen, 28 Nov.). Other CMs expressed similar concerns: “The 0.25% false positive rate sounds impressive until you consider more intuitive measures of performance. If the bot makes 2,500 reverts per day, that’s 2,410 good reverts and 90 false positives per day. While that may seem like great progress from a data perspective, Wikipedia is also a community of editors” (Stepheng3, 28 Nov.). Under pressure from the requests of multiple CMs to reduce the false-positive rate, one developer made adaptations that reduced it to 0.1%.

Flagging: Whereas scoping and adapting focused on adjusting the bot’s behavior to improve consistency with community expectations and prevent harm to the coproduction

process, *flagging*, introduced during the first trial, allowed CMs to question the bot's reversion decisions and request human review of those decisions (C6). Unlike scoping and adapting, flagging did not lead merely to one-time adjustments to the bot. Instead, it enabled CMs to flag and review the bot's questionable reversions on an ongoing basis.

Flagging enabled CMs to highlight erroneous decisions made by ClueBot. Flagging became relevant when developers implemented a feature that allowed requests for human assessment of the bot's decisions in cases of uncertainty. For instance, after the bot flagged several valid edits as vandalism, a user raised the following concern: "I edited the *Greatest Hits So Far* redirecting page to a disambiguation page to include both *The Greatest Hits, So Far* and *Greatest Hits ... So Far!!!* since it is an upcoming release. Yet, it was reverted and I received a vandalism warning. The same thing happened when I corrected the UK edition track listing, which is currently incorrect" (Imjayyy, 3 Nov.). In response, developers introduced special handling codes to prevent false positives, particularly for content that had weak representation in the dataset: "We will add these reported false positives to the dataset and add special handling code for these" (Crispy1989, 3 Nov.).

Human editors increasingly reported false positives for edits characterized by ambiguity. As one editor suggested, "In the absence of a better solution, articles [characterized by ambiguity] should be removed from the 'vandalism' dataset" (Peter Karlsen, 2 Nov.). These reports prompted developers to introduce disambiguation tags, which allowed editors to flag edits that could not be easily categorized as either vandalistic or constructive. This tagging system enabled the community to collectively decide on the appropriate course of action: "I also added special handling for disambiguation tags. Testing the fixes now; bot will be restarted with updates soon" (Crispy1989, 3 Nov.).

Escalating, which resulted in one adaptation to the bot (C13), is an authority-negotiation design move that went beyond flagging in that it provided a way for CMs to override the bot's

decisions when necessary. Escalating allowed CMs to broadcast flagged bot reversions to the broader community and introduced a protocol that allowed the community to vote on whether a bot's questionable or ambiguous edits should be overturned, thereby reinforcing the bot's learning.

The community dialogue about the need for such a voting mechanism was a direct response to the increasing disagreement among CMs and developers over ClueBot's reversions and how they were being handled. For example, one CM suggested that human judgment was necessary when blanking (i.e., the removal of all or a substantial part of a page's content) occurred:

“While content removal on BLPs [biographies of living persons] can constitute vandalism, human judgement is required to determine whether this is actually the case; the content may have blatantly violated the sourcing or neutrality requirements of the policy” (Peter Karlsen, 26 Oct.).

Moreover, the need for a consensus-building mechanism was increasingly emphasized in cases of borderline vandalism, that is, good-faith edits that the bot classified as vandalism because their quality was poor. Many editors felt that, in such cases, the community consensus should avoid discouraging human editors, especially those with little experience: “The bot undid an IP user's good faith edit, a perfectly fine insertion of a word for clarification. I have reverted back and intend to post on the IP talk page that it was a mistake. I realize it might be harder to distinguish between such edits and some forms of vandalism, but I imagine minor edits of this type are common and we don't want to deter casual editors” (ChiZeroOne, 18 Nov.). Furthermore, some CMs disliked suggestions to leave edits unconsidered if CMs disagreed on a borderline edit, requesting wider community input for the assessment of vandalism.

Motivated by these arguments, ClueBot's developers implemented a voting mechanism that allowed human editors to classify potential occurrences of vandalism, that is, to vote on

whether an edit should be classified as vandalism, whether it should be classified as constructive, or whether the decision should be skipped (e.g., when editors could not make an informed decision). The discussion then turned to the questions of when a decision could be made based on these votes and the extent to which editors should be able to see the votes of other editors. For instance, editors initially expressed an interest in receiving information about how other editors categorized edits, mainly because they wanted to know which edits the community generally considered vandalistic in order to improve their own classification skills (e.g., “If there is any disagreement with another reviewer about any of my classifications then I’d appreciate learning about it, if only to improve my own performance” [Philip_Trueman, 6 Nov.]) or in order to avoid misunderstandings (e.g., “I didn’t really like the suggestion I saw somewhere that if two reviewers disagree then the edit will be dropped from the dataset—surely that is a recipe for blunting the sensitivity of the bot? If that is the case, then I wonder what is happening to all those comments I’ve placed on difficult edits. ... We’d all learn something, the reviewers as well as the bot” [Philip_Trueman, 12 Nov.]).

Eventually, ClueBot’s developers responded by supplementing the voting mechanisms with rules that determined when a decision about a particular edit was reached: “For discarding edits where there’s some disagreement, we’ve decided to change this to a scheme where every edit is always classified as something (vandalism, constructive, or skip), and that the classification that is used must have at least 3x the votes as any other classification” (Crispy1989, 13 Nov.).

Requesting, which was observed during the first and second trials, involved requests for changes to ClueBot that facilitated CMs’ direct feedback to the bot and their involvement in reinforcing the bot’s learning. Requesting resulted primarily in adaptations to the ClueBot interface that solicited community engagement in the training and improvement of the bot; such adaptations aimed to reduce conflict caused by the bot’s interface, which was not

conducive to fostering community contributions and transparency. In total, requesting resulted in four changes to the bot (C10, C12, C14, C15); these changes related mainly to the bot's interface.

Changes implemented through requesting enabled all the editors to participate in the monitoring of the bot by allowing them to easily report and correct errors. On several occasions, the developers asked CMs to help them fix and train ClueBot, and one developer described the use of the interface for this purpose: "The interface can be found at this link. ... If you'd like to help out, please follow the link and go to the signup section. Help is needed, and greatly appreciated!" (Crispy1989, 3 Nov.). However, problems with the engagement of editors quickly arose. For example, some editors received unintelligible messages after registering to contribute to the development of ClueBot, and some had to request better instructions for using the interface to categorize disputed edits: "Is there any, umm, help or documentation for this interface?" (Philip Trueman, 5 Nov.). Responding very quickly, a ClueBot developer made the user interface, and especially the messages that notified users of a successful registration, clearer and more intuitive. The interface, however, was not only improved to increase the participation of editors but also adjusted to reassure CMs when they received feedback on difficult edits: "Though it would be nice if 30% of edits for review were not my own ... To encourage wider participation, include some stats (e.g., 'after the nth review, bot accuracy has improved 5%' or whatever it is)" (Rjwilmsi, 13 Nov.).

The developers also proposed introducing greater transparency into the classification of edits by making discrepancies between the classifications of different editors visible and by providing a basis for discussions in which editors could attempt to resolve such discrepancies or articulate alternative views: "For getting feedback on difficult edits, we've discussed ways to do this, and it may be possible to set something up. ... The idea is to allow users to view a

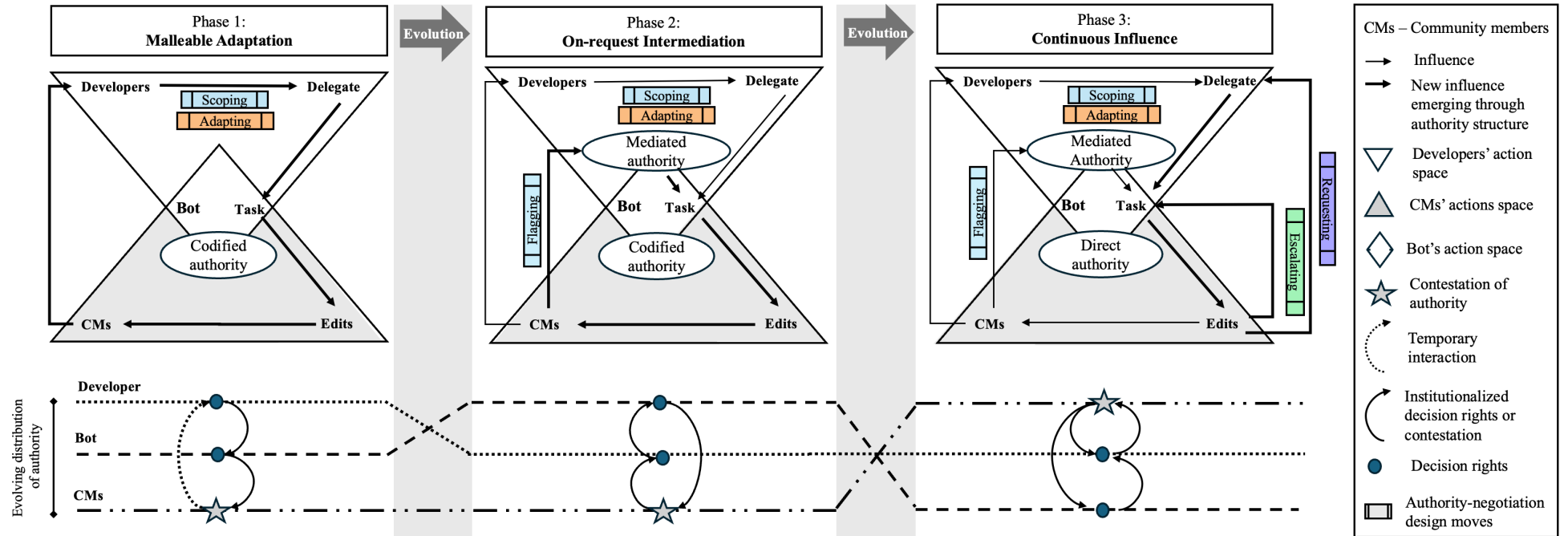
list of all edits they've classified, that others have classified differently, and allow them to view and add more comments and change their existing vote" (Crispy1989, 13 Nov.).

Additionally, developers implemented vote visibility: "One of the important asked-for improvements is that users can now view what others have voted on edits they've already reviewed by clicking on the counter in the top-right corner, and potentially change your own vote in retrospect. Note that you cannot view what others have voted before voting yourself—this is to prevent any prior bias" (Crispy1989, 22 Nov.).

A Theory of How AI-Human Authority is Dynamically Redistributed

Building on the identification of authority-negotiation design moves, we show how these moves accumulated into evolving structures of authority that redefined interactions among actors and progressively increased the influence of CMs over developers and the bot's authority and impacted the bot's task execution. This process was marked by a recurring process of contestation and adaption of bot authority by CMs and produced three dynamics: the redistribution of authority from the bot to developers and CMs, the reshaping of influence from temporary negotiation during the approval process to institutionalized influence of CMs on both developers and the bot, and the evolution of bot authority enacted in response to these changes, producing new sites of contestation. Next, we summarize these dynamics, which emerged over three distinct phases (also see Figure 4).

Figure 4. *Evolution of Authority-Negotiation Design Moves and Redistribution of Human and Bot Authority.*



During the first phase, *malleable adaptation*, authority-negotiation design moves took the form of scoping and adapting, both of which involved changes to the bot's code and produced what we term codified authority. Codified authority arose when the bot's permissible domain and decision parameters were reset through rules and thresholds embedded in software (e.g., excluding certain namespaces, whitelisting experienced editors, lowering false-positive rates), thereby constraining the bot's action space and structuring how authority is exercised over editors. These changes durably limited the bot's reach but also created new points of contestation. For example, once newcomers were protected from bot reverts by removing prior-contributions that contained minor editing mistakes from the training data (C1), experienced editors demanded parallel safeguards, prompting additional scoping (C5). Analytically, these adjustments marked the emergence of codified authority: boundaries and parameters were fixed in code but remained open to challenge and further refinement through authority-negotiation design moves.

This phase captures a configuration in which authority was codified in software yet remained open to renegotiation, expanding opportunities for CMs to influence developers. Such influence, however, was confined to the approval process. CMs could persuade developers through argument and evidence, but developers retained ultimate authority over code changes, and the bot retained authority over edit reversions. Authority was therefore codified in software, embedding lasting constraints on the bot's scope and decision parameters, while CMs' influence remained temporary and contingent on their ability to convince developers under BAG oversight.

The second phase, *on-request intermediation*, was characterized by the negotiation move of flagging, creating what we call mediated authority. Mediated authority arose when CMs could not directly overturn a bot decision but instead redirected contested decisions to developers through a formal mechanism. Flagging thus enabled developers to exercise authority over

individual bot outcomes, without altering the overall scope, in response to CM concerns. In our analysis, this move marked the implementation of more durable forms of contestation: Flagging gave CMs a persistent means of activating case-bounded developer oversight and mediation. Mediated authority is thus a configuration in which CMs shaped outcomes indirectly by involving developers in the evaluative process.

In this phase, authority shifted as developers gained decision rights they would not otherwise have assumed absent flagging. At the same time, flagging institutionalized a structure for sustained contestation, giving CMs a persistent channel to activate developer oversight whenever the bot's authority was disputed. This arrangement redistributed authority by strengthening developers' role in day-to-day decisions and by granting CMs a formal mechanism to trigger such interventions. CMs' influence also shifted. First, flagging provided a more formalized way to challenge questionable reversions through an interface rather than through ad hoc talk-page debates. Second, it afforded CMs a mechanism that could be invoked selectively, enabling contestation to occur on demand.

The third phase, *continuous influence*, emerged through the authority-negotiation design moves of escalating and requesting, introducing what we term direct authority. Direct authority arose when CMs gained an immediate and formalized means to intervene in bot decisions or demand changes without relying on negotiation or developer mediation. Escalating enabled CMs to overturn contested bot decisions collectively through voting mechanisms, while requesting allowed them to propose and secure interface changes that shaped how both developers and the bot operated. In our analysis, these moves marked the point at which CMs no longer influenced authority through negotiation or case-bound mediation requests but were able to make use of escalating and requesting authority-negotiation design moves to influence or override the bot's decisions on an ongoing basis without developer mediation.

In this phase, CMs established enduring structures that embedded their influence directly into the bot's operation and the developers' decision space. Authority was thus eventually redistributed across all three actors: the bot continued to exercise authority over human edits, developers retained control over code, and CMs acquired lasting, formalized channels to shape both. This phase institutionalized continuous influence, as recursive interactions among CMs, developers, and the bot created stable yet dynamic arrangements for adjusting authority.

In sum, our findings show how authority-negotiation design moves crystallized into structures of codified, mediated, and direct authority that progressively reconfigured the influence of CMs, developers, and the bot. Across the three phases—malleable adaptation, on-request intermediation, and continuous influence—authority was contested, redistributed, and reenacted in ways that expanded the scope of CMs' influence from temporary negotiation within the approval process to institutionalized, ongoing involvement in developers' and the bot's decision-making.

DISCUSSION

We began with the observation that the IS literature knows little about how authority is reshaped once it is delegated to a supervisory bot. In the case of ClueBot at Wikipedia, this reshaping unfolded through a gradual increase of CMs' influence over both developers and the bot over a 24-day time frame, realized through the implementation of authority-negotiation design moves. What started as episodic contestation within a bounded approval process evolved into changes in structures of authority, through which CMs gained sustained influence over how bot authority was enacted. The findings of this research have several notable theoretical implications.

Theoretical Implications

First, this research contributes to the growing IS literature on the delegation of decision rights to AI-based, autonomous bots by theorizing how authority is *dynamic*, rather than static,

in such arrangements. Research on this phenomenon has begun to draw attention to the implications of delegating authority to intelligent technologies such as AI-based bots (e.g., Fügener et al., 2022; Jussupow et al., 2021). While such work has advanced our understanding of the impacts of delegating authority to intelligent technologies, much of it either implicitly assumes that authority remains static or neglects to examine what happens to authority once it is in place. Through the lens of design moves, we show how authority evolves after it is delegated to the bot by being progressively reshaped through authority-negotiation design moves. In our case study, these moves—scoping, adapting, flagging, escalating, and requesting—materialized in concrete design changes negotiated by CMs and implemented by developers. Each move was crystallized into structures of codified, mediated, or direct authority, progressively expanding the scope and persistence of CMs’ influence. In this way, the contestation of bot decisions provided a mechanism for translating community concerns into lasting institutional arrangements. The authority-negotiation design moves surfaced in this research complement the design-oriented approaches to addressing authority vested in intelligent technologies (e.g., Kane et al., 2020).

Second, we contribute to the literature on delegation of decision rights to AI-based bots by theorizing how authority is redistributed among relevant actors in such arrangements. The redistribution of authority is not something that the current discourse on delegation has focused on, yet it is essential to understanding how to facilitate constructive work arrangements between humans and bots. Our analysis reveals that the authority-negotiation design moves that we identify can shift the locus of performativity of authority. In the Wikipedia case, authority was initially performed through the code specified by developers and enacted by the bot. As contestation emerged, however, CMs increasingly became central actors in shaping authority enactment. The developers’ role shifted in response to this contestation. First developers distributed authority by transferring it to the bot. Second, they mediated authority

when responding to community discourse or reacting to flagged bot decisions. Third, they modified authority structures by implementing SAFER authority-negotiation design moves that reshaped authority permanently, allowing CMs to overrule bot decisions themselves or request new features from developers outside of the bot approval process. In this sense, developers moved from acting as distributors of authority to co-constructors of evolving authority arrangements together with CMs. The locus of performativity thus shifted from developer-bot interactions to CM-developer dynamics mediated by flagging to a tripartite configuration in which CMs, developers, and the bot collectively shaped authority enactment. This redistribution was not a one-time reallocation of decision rights but a dynamic process in which each new structure of authority created further sites of contestation and adaptation. This finding draws attention to the importance of considering the dynamics of who enacts authority and in what way in delegation arrangements.

Third, this research uncovers the gradual institutionalization of influence as one means through which those who are subject to bot-delegated authority can reclaim their own authority. Prior research has not explicitly accounted for how this occurs. In our study, early contestation allowed CMs to voice concerns and persuade developers in bounded ways, but such influence was contingent and temporary. As mediated and direct authority structures emerged, CMs gained formalized and durable mechanisms to shape both bot outcomes and developer decisions. In doing so, their influence became continuous, embedded in the governance of the bot beyond the approval process. This progression illustrates how authority, once delegated to AI-based systems, can evolve from discretionary developer control to institutionalized arrangements in which those subject to authority are empowered to shape it.

Together, these findings illuminate the social dynamics through which authority is reshaped after delegation, and they carry important implications for IS research on delegation and design. These insights extend IS research on delegation by shifting attention from initial

allocation decisions to the dynamic practices through which authority is redistributed, reconfigured, and institutionalized over time. In doing so, we move beyond the dominant view of delegation to bots as a one-time transfer of decision rights, showing instead that authority evolves as those subject to bots contest, negotiate, and embed new structures of influence. This highlights that delegation to bots differs fundamentally from delegation to humans: bots cannot themselves negotiate authority, but their authority is continually reshaped by the practices of those they supervise, mediated through developers and code. Our findings also suggest that while design is important for shaping how authority is initially delegated, design-oriented approaches underestimate the extent to which authority is enacted and reshaped through social processes and evolving structures. Aligned with the performative view of authority, our study shows that it is not only technical design choices that matter but also the ongoing contestation and institutionalization of practices that determine how authority unfolds in use.

Practical Implications

Our findings offer practical guidance to managers and developers navigating human-bot collaboration. We investigated how authority evolved after delegation in the context of an open production community, not because bots operate exclusively in such settings but because the participatory, meritocratic, and transparent nature of such settings provides a uniquely suited context for tracing how the authority of bots is challenged and reshaped and observing how authority is redistributed among developers, bots, and other stakeholders over time.

However, the implications of our findings extend beyond this setting. Many contemporary organizations operating with low or fluid formal hierarchies, such as startups, agile teams, and platform organizations, embed bots into shared workflows. In such environments, authority is also negotiated in practice rather than imposed unilaterally. Similarly, decentralized autonomous organizations are sociotechnical systems in which authority is increasingly embedded in code yet remains subject to collective oversight. The configurations of authority

we identify represent a conceptual lens that illustrates how practitioners can recognize and respond to emerging shifts in influence and control among humans and bots.

In view of these diverse contexts, we ask: How can organizations implement authority-negotiation design moves as actionable mechanisms for employees? This study identifies five authority-negotiation design moves, which we term SAFER—scoping, adapting, flagging, escalating, and requesting—that enabled CMs to reshape the distribution of authority following the introduction of ClueBot. These moves offer guidance for the design of AI-based bots: They illustrate how human actors can retain influence over automated systems and how developers can proactively embed associated mechanisms that support the reliability and contestability of bot decisions after the initial delegation of authority.

We propose that SAFER can be applied in the design of various kinds of AI-based bots. Although our identification of the SAFER moves resulted from a study of ClueBot, which was based on a form of AI that is far less advanced than contemporary forms of generative AI (such as ChatGPT), these moves remain relevant to more sophisticated forms of human-AI collaboration on digital platforms. For instance, the design of ChatGPT incorporates several SAFER-like mechanisms. Users can engage in *flagging* for internal review responses that seem inappropriate or incorrect (OpenAI, 2024). Although ChatGPT lacks formal *escalating* mechanisms, users can make *requests* by asking for specific features or improvements in community forums, which OpenAI monitors. Although these mechanisms are handled internally by employees of OpenAI rather than in a community-driven manner, as on Wikipedia, they can still allow users to shape and contest the bot’s behavior within the constraints of the system. Organizations could explore implementing SAFER in a variety of other AI tools as well. For example, a customer-service bot could incorporate *scoping* processes that would allow supervisors to restrict a bot’s authority to certain transaction types,

flagging systems that would enable workers to mark problematic bot decisions for review, and *escalating* protocols that would route complex cases to human managers.

This practical applicability leads to two important concerns regarding the use of the SAFER by developers and technology users. First, it should be noted that we derived the SAFER authority-negotiation design moves from a community form of production and management. However, in for-profit contexts, hierarchical structures may limit individuals' opportunities to influence the distribution of authority. Therefore, we recommend that developers, along with other parties with authority over the initial delegation of tasks to bots, implement SAFER mechanisms proactively. The case of ChatGPT shows that even in forms of engagement that are less democratic than Wikipedia's community approach, SAFER-like mechanisms can give humans who interact with bots a stronger sense of authority.

Second, we argue that the impact of the SAFER moves may be more significant for individuals who encounter AI-based bots as a part of their work routines (regardless of whether they are volunteers or paid workers) than for those who choose to use such bots voluntarily. For tools like ChatGPT, humans have initial authority over task delegation and can even decide not to engage with a bot. Although it is advisable to implement SAFER moves in such cases, it is important to recognize that their impact may be less pronounced in contexts of voluntary use.

Finally, we emphasize that in some scenarios—especially in highly regulated industries that limit robust community involvement—a full implementation of SAFER moves may be inappropriate. In regulated industries, it may be prudent for developers to explore the implementation of SAFER in the early phases of the design and development of AI-based bots. Doing so could help them develop a context-specific means of identifying appropriate and inappropriate bot actions. Moreover, developers may want to consider the unintended consequences of implementing measures that allow humans to directly influence bots and their

decision-making. For example, malicious actors could engage in the practice of escalating and influencing voting to silence authentic information provided by bots or to bias the future behavior of bots. Because these considerations are beyond the scope of this paper and require further elaboration, we leave them to future explorations of the optimal use of the SAFER-like mechanisms for the development and design of AI-based bots in the workplace.

Limitations and Future Research

Like all studies, ours has limitations that should be addressed in future research. First, we drew our results from a study of a single case, that is, the deployment of Wikipedia's AI-based antivandalism bot ClueBot. Although a single-site case study inherently has limited generalizability, we believe that important elements of our findings are relevant to many settings that use AI-based technologies. Our narrative analysis resulted in a temporal tracing of how the authority conferred on bots was navigated by central stakeholders, namely CMs. This tracing is based on our analysis of rich data on bot changes, negotiations, authority shifts, and the linkages between them. Unfortunately, such rich analysis cannot be economically achieved using large samples (Langley, 1999). However, our analysis uses a grounded theory approach to go beyond merely idiosyncratic storytelling (Langley, 1999), allowing us to develop a deep understanding of the evolution of authority redistribution and consequent configurations of authority redistribution between delegators, delegees, and supervised parties. Although this theory is provisional, it has the potential to be developed into a general theory.

Future research could elaborate on authority configurations by evaluating their sustained impact on the execution of structural, dispositional, and relational power in various contexts and by validating the presence of authority-negotiation design moves in the context of other AI-based bots used by Wikipedia or by other organizations that perform evaluative functions. In particular, we believe that exploring the configurations of authority redistribution in generative-AI applications is a particularly fruitful avenue for future research. This research

could extend beyond Wikipedia to examine other open co-development communities or organizational settings. For example, scholars could partner with companies like OpenAI to determine how to structure interactions between private developers and bots in the context of the fine-tuning of large language models. Such work could offer insights into the handling of edge cases and AI hallucinations (OpenAI, 2024).

Second, although the evolution of authority redistribution is relevant in various scenarios, such as for-profit contexts and organizational contexts in which humans work alongside AI-based bots, we acknowledge that our findings are derived from a context defined by open-community forms of production and management. As a result, our findings require replication in hierarchical settings, such as large organizations, in which human involvement in establishing rules for AI-based bots may be considerably different. For example, in for-profit organizations, a centralized committee may establish rules for task delegation to AI-based bots and their authority without taking employees' preferences into account. Attempts to replicate our findings should also consider industry-specific requirements, such as regulatory requirements or stakeholder demands, that might preclude the implementation of certain mechanisms. Future research could thus focus on elucidating the conditions conducive to a process of redistributing authority, considering industry-specific requirements, organizational structures, and organizational cultures to determine how power is distributed within and outside organizations. Such guidelines could provide crucial guidance on how to distribute authority between humans and AI in such organizations.

Third, while the discussions on Wikipedia's talk pages provide extremely rich and unfiltered documentation of CMs' concerns and the responses of the bot's developers, both of which shaped the distribution of authority between Wikipedia's CMs and its AI-based bot, there are some analytical drawbacks associated with this case. Because ClueBot was introduced in 2010, the data are not recent; thus, our conclusions require careful interpretation.

Although the data may be dated, we argue that this does not compromise the quality of our insights or of our emerging theory. We derived our results based on the introduction of an AI-based bot that occurred when the capabilities of AI were much less developed than they are now. However, although the capabilities of AI have grown—for instance, non-supervisory bots are more sophisticated when powered by generative AI—our findings continue to be relevant. Indeed, the central questions of this study—for example, when and how much authority should be delegated to bots, how can stakeholders negotiate this authority after it is delegated, and how is this authority reshaped over time—are no less important today than they were a decade ago.

Fourth, although our study focuses primarily on the human delegation of tasks to AI-based bots, we acknowledge that the capabilities of AI now extend beyond this unidirectional relationship. Inversion, which is the ability of AI to delegate tasks to humans (Baird & Maruping, 2021; Fügener et al., 2022), is an important phenomenon that our theorizing does not explicitly include. We recognize that the redistribution of authority between humans and AI-based bots over time can account for such scenarios. The ability of AI to transfer tasks back to humans may become a significant component of this evolving configuration of authority distribution. This means that as the collaboration between humans and AI-based bots progresses, the authority of AI may expand to include bots delegating tasks to humans.

Finally, it is necessary to reflect on the exclusionary openness of Wikipedia. Scholars have shown that contributing to Wikipedia requires technical skills that are not easily acquired by newcomers (Dobusch et al., 2019). Additionally, Wikipedia CMs have generated many rules and codes of conduct that are not transparent and are enforced by experienced editors to further their own interests and agendas (Jemielniak, 2014). We therefore need to acknowledge that our analysis relies on a community discourse carried out by a small group of experienced CMs and that our study therefore does not include voices that are less technologically savvy or otherwise

marginalized. Furthermore, the supervision of the test phases by a group of experts, the BAG, put pressure on developers to adapt the bot and thereby redistribute authority, changes that otherwise might not have occurred. Therefore, our final recommendation for future research is to evaluate the effectiveness of the authority-negotiation design moves for adapting the configuration of structural, relational, and dispositional power among larger, more diverse groups of humans who will be asked to work alongside AI-based bots that hold authority. Online or field experiments could be instrumental in assessing how mechanisms that (re)distribute authority between humans and AI build trust or create suspicion and how they might shape the quality of human-bot collaboration.

CONCLUSION

This study examined how authority evolves once delegated to AI-based bots. While prior IS research has emphasized delegation as a decision to transfer tasks or decision rights, we analyzed a different configuration: bots assigned evaluative functions that authorize them to judge and intervene in the work of others. In such cases, authority is not static but subject to contestation by those supervised, raising the question of how authority is reshaped over time.

Using the case of ClueBot on Wikipedia, we showed how authority initially centralized in developers and enacted by the bot was progressively redistributed and reconfigured through CMs interventions. Through five authority-negotiation design moves, CMs influenced both the boundaries of the bot's evaluative scope and how developers exercised formal authority. These moves crystallized into structures of codified, mediated, and direct authority, each altering how outcomes were decided and by whom. Over time, what began as temporary influence during the approval process evolved into institutionalized arrangements that embedded CMs' continuous influence on both developers and the bot.

By placing authority rather than delegation at the center of analysis, our study shows that although the enactment of bot authority can change over time, the reshaping of authority is

driven by humans. Community members contest the bot's actions, and developers translate this contestation into authority-negotiation design moves that adjust the rules, thresholds, and parameters through which the bot operates. These changes reconfigure the structures and influence through which authority is exercised and redistribute authority across CMs, developers, and the bot, opening new sites of contestation. The SAFER authority-negotiation design moves capture this human process and show how authority delegated to bots is continually reshaped through recursive interactions between developers and CMs in response to the bot's decision-making.

Practically, these findings highlight that delegating evaluative tasks to bots requires more than careful design when the bot is deployed. Lasting legitimacy depends on creating sociotechnical arrangements that enable contestation, feedback, and adjustment as bots enact authority in practice. Structures that allow those subject to bot authority to influence and renegotiate it are essential if organizations are to balance efficiency with fairness and sustain participation in AI-mediated environments.

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